



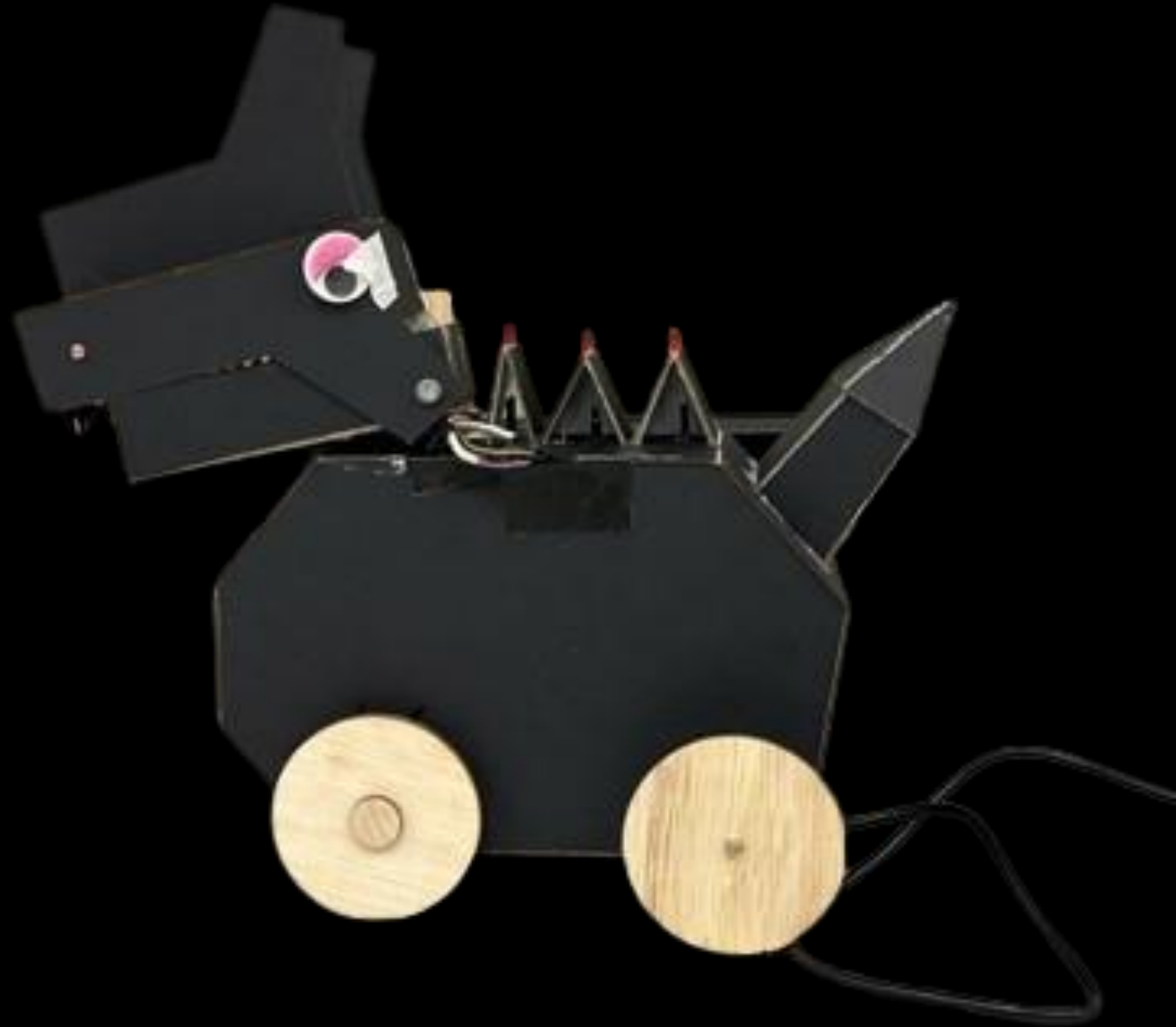
University of Southampton MALAYSIA

Group 10

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The Arduino Powered Dinosaur

Rexduino – a combination of timeless allure of the T-rex with the advanced capabilities of modern society technology. This interactive robotic dinosaur brings life to the imagination through dynamic movements, sensory responses, and engaging interactions.



The Inspiration Behind Rexduino

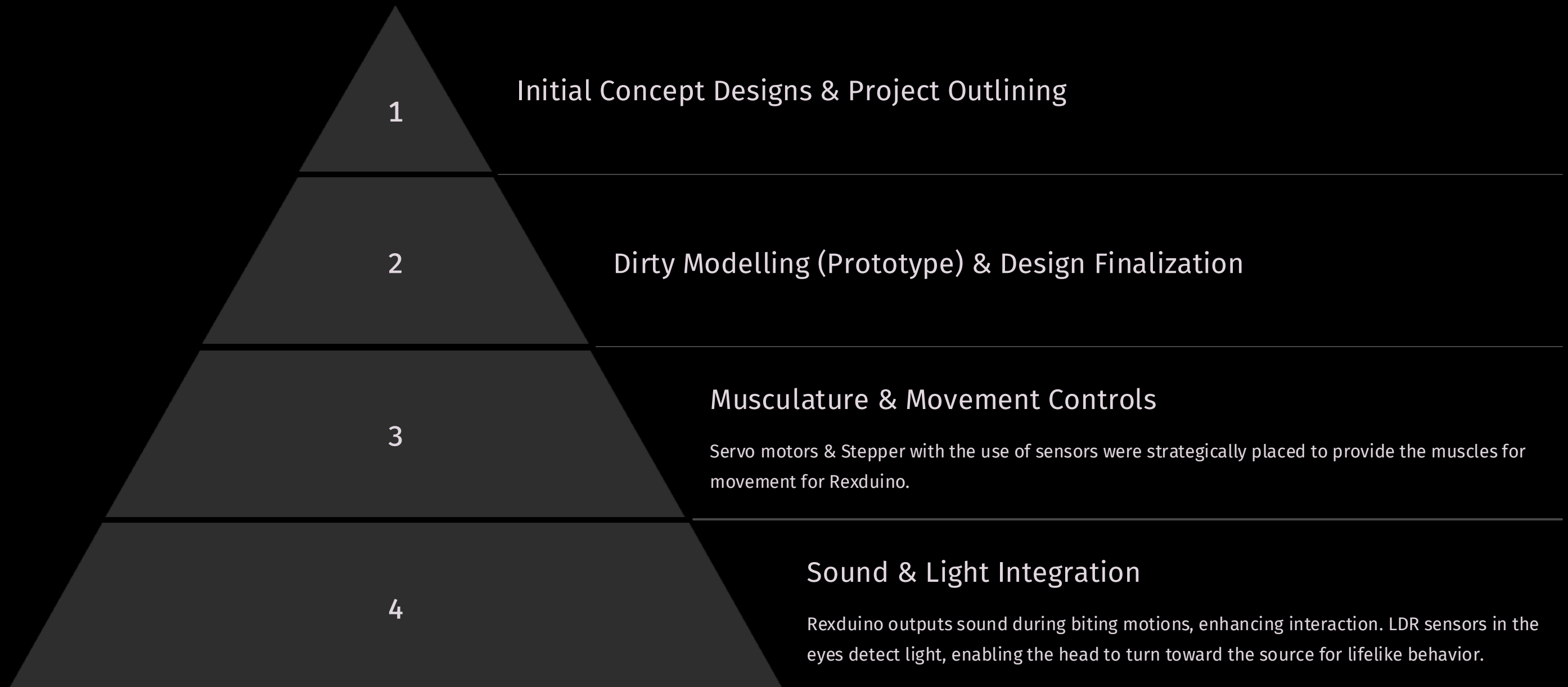
A Childhood Dream

Rexduino was born from a simple yet powerful desire: to showcase the wonder of dinosaurs to life. The project draws inspiration from the awe-inspiring T-rex that once roamed the earth, sparking imaginations for generations.

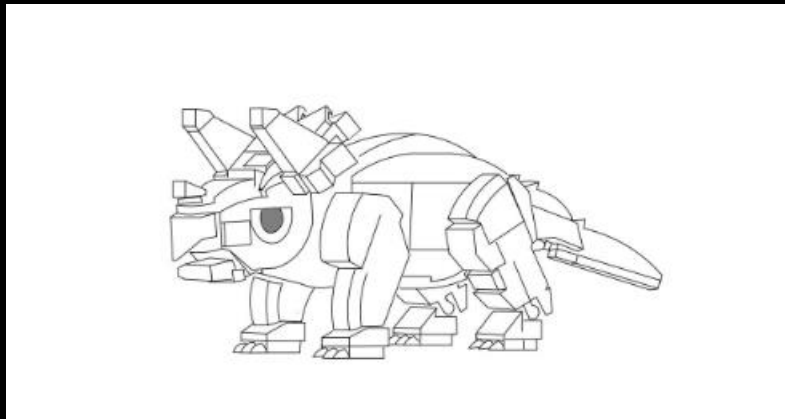
Bridging the Gap

The project aims to bridge the gap between childhood fascination and the cutting-edge world of robotics. It demonstrates that even complex creations can be brought to life with accessible technology and creative ingenuity.

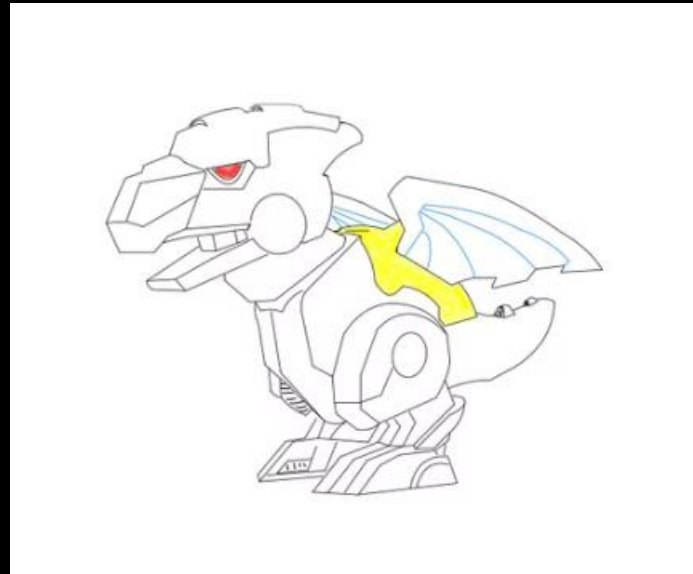
Designing the Mechanical Dinosaur



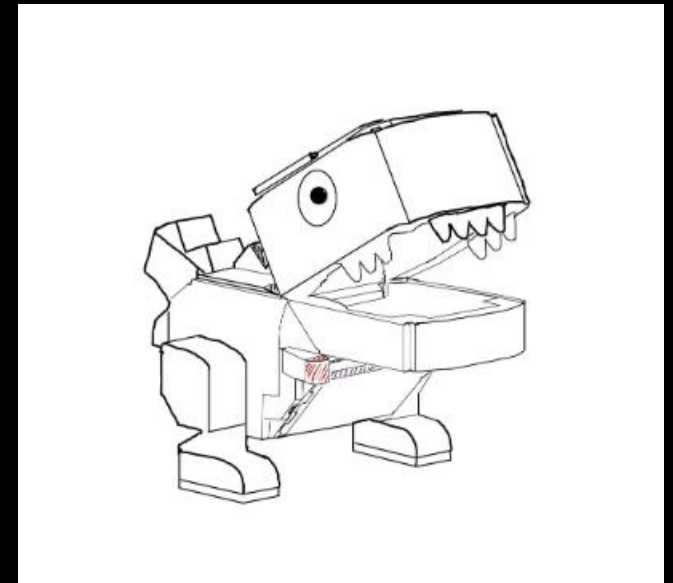
Initial Concept Designs



Triduinodroid



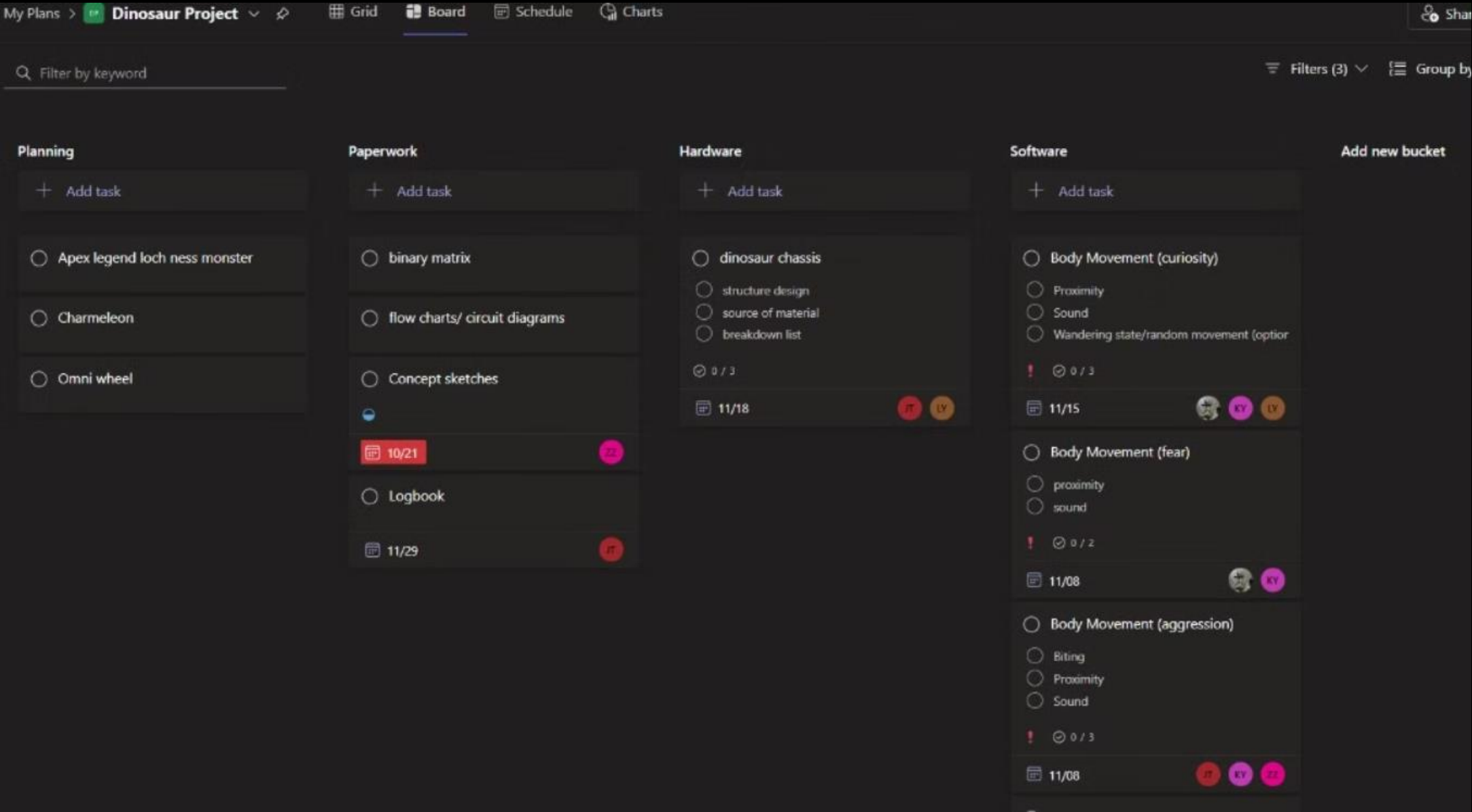
Pterodruinosaur



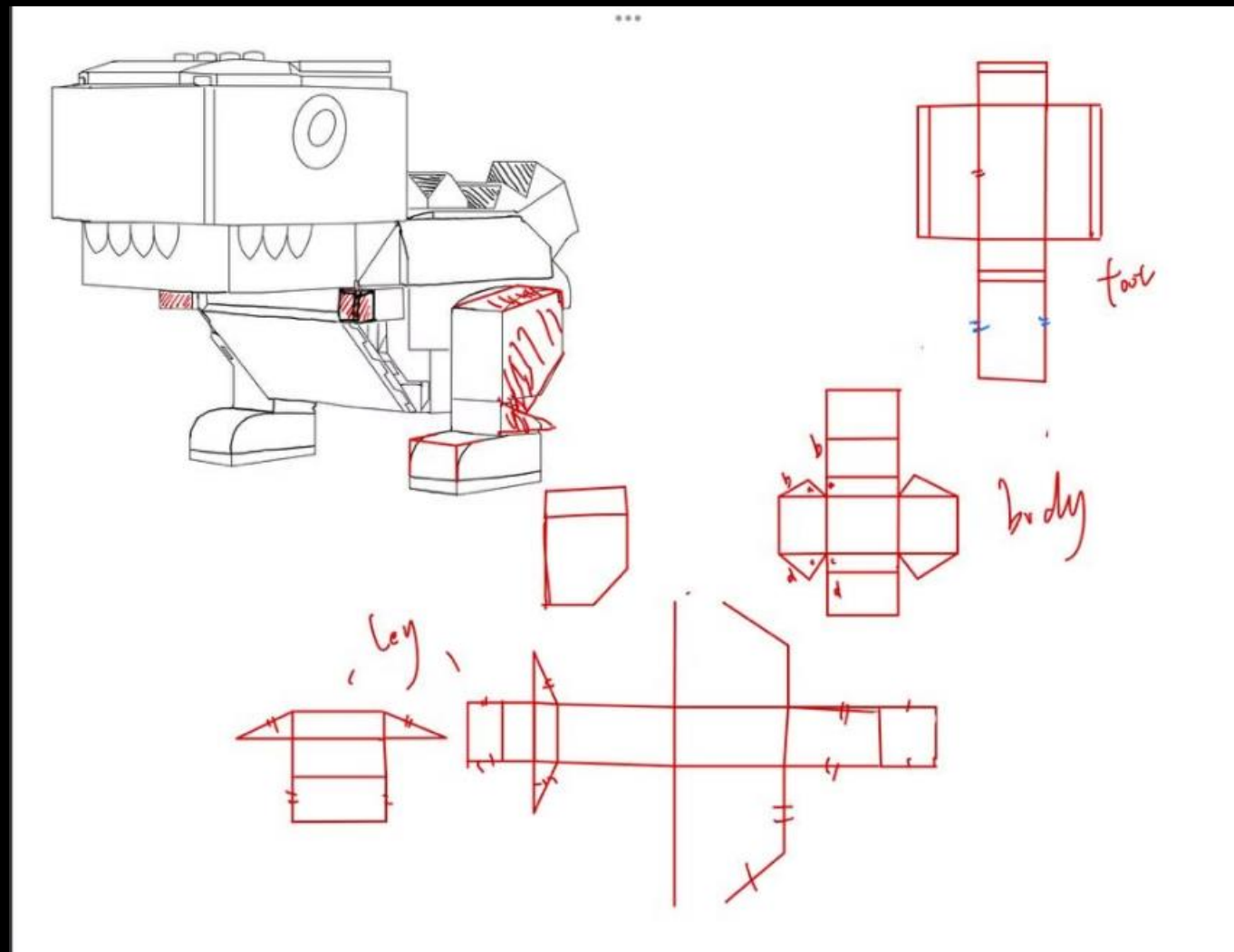
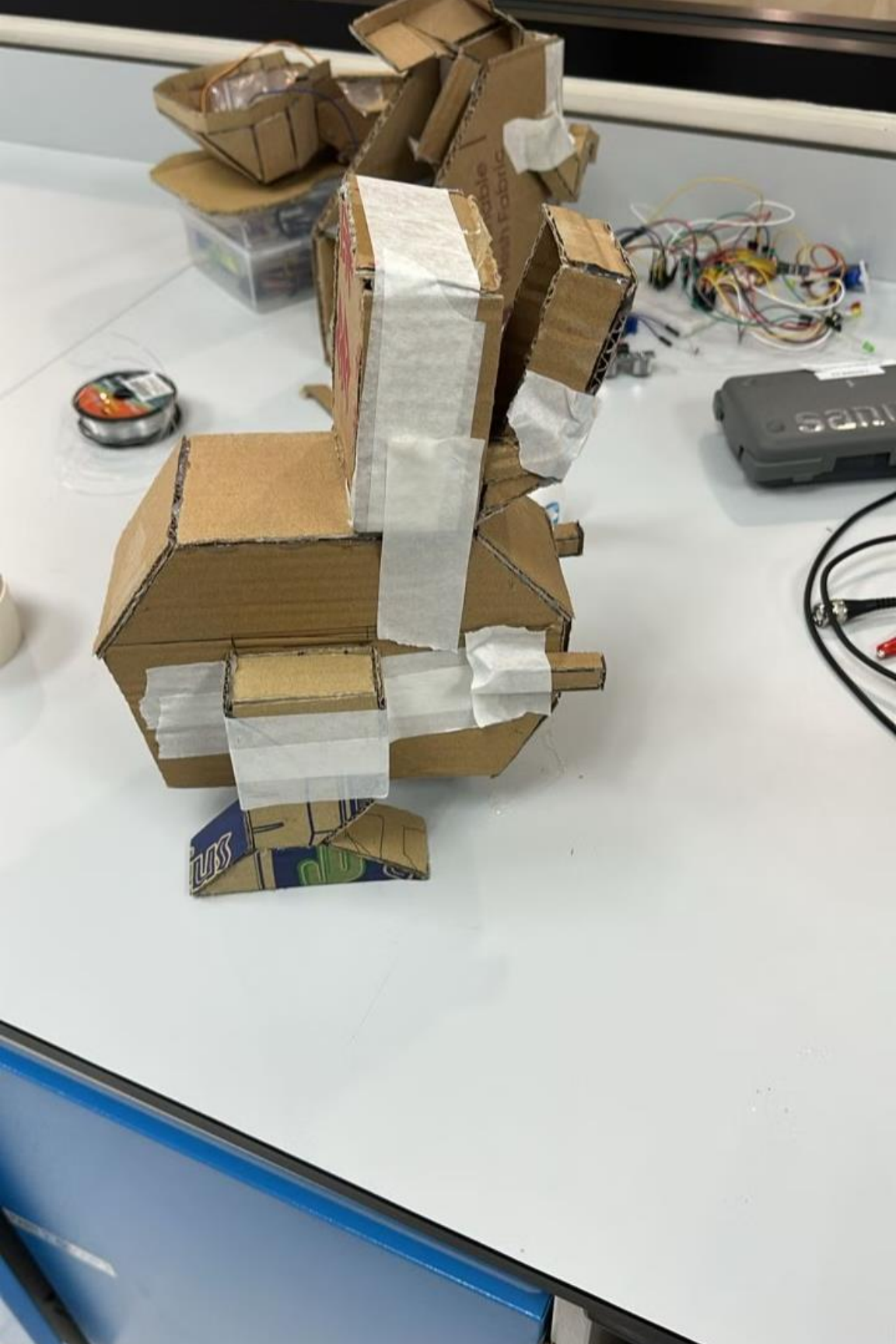
Rexduino

Project Outlining

Utilizing MS Team to plan the project accordingly



Dirty Modelling (Prototyping)



Integrating Arduino for Interactivity

1

The Brain: Arduino Uno

The Arduino Uno serves as the central processing unit, controlling Rexduino's actions and reactions.

2

Sensory Inputs:

HC-SR04 Ultrasonic Sensor: Detects distance, triggering curiosity or fear responses.

LDR (Light Dependent Resistor): Detects light intensity changes.

3

Output Control:

1) Stepper Motor: Controlled by the HC-SR04 sensor, moves Rexduino forward and backward based on detected distance.

2) Servomotors:

- **Neck:** Turns the head towards light sources detected by the LDR sensor.
- **Mouth:** Simulates biting when the LDR is covered.

3) LED Light (Backbone): Illuminates when the mouth is closed and during biting, providing visual feedback.

4) Buzzer: Emits a sound during biting actions, adding auditory feedback.

Bringing Rexduino to Life: Coding and Electronics

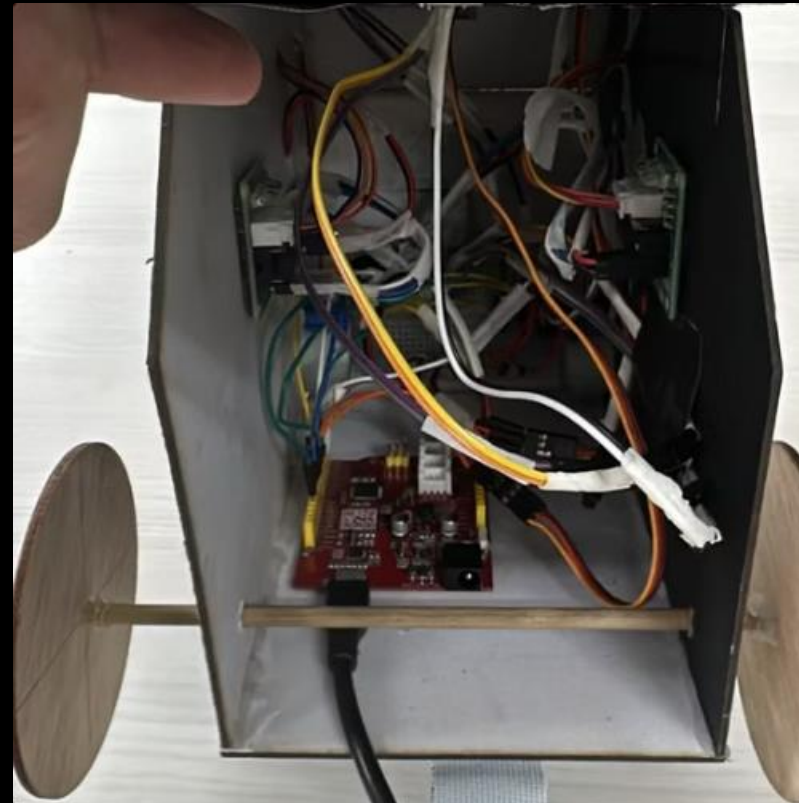
```

1 //Arduino Servo.h
2 #include <Servo.h>
3
4 Servo mouth_servo;
5 Servo neck_servo;
6
7 // Pin definitions
8 #define photo_resistor1 A1
9 #define photo_resistor2 A2
10 #define LDR_A0
11 #define echoPin 2
12 #define trigPin 3
13 #define redID 11
14 #define C01 342 //I Dc_
15 #define B01 324 //I Rr_
16 #define F01 330 //I M_
17 #define F01 349 //A Pa_
18 #define G01 392 //S Sol_
19 #define G01 415 //WS Sol_
20 #define A01 448 //Lc La_
21 #define B01 494 //Tt Ss_
22 #define C01 519 //I Dc_h
23 #define D01 547 //I Rr_h
24 #define E01 632 //R2 Rss_h
25 #define F01 650 //I M_h
26 #define F01 698 //A Pa_h
27 #define G01 784 //S Sol_h
28 #define A01 808 //Lc La_h
29 #define B01 908 //Tt Ss_h
30 #define C01 900 //停止用 pause
31
32 int angle_mouth = 90;
33 int angle_neck = 90;
34 int LDR_left;
35 int LDR_right;
36 int mouth_R;
37 int r_red = 255;
38 int counter_mouth = 0;
39 int D=0;
40 int scale[]={C01,D02,C03,D02,F01,B01,D01,C01,A01,G01,F01,A01,B01}; //input
41 int time[]={D,1,1,1,1,1,1,1,1,1,1,1};
42 int length;
43
44 // Motor setup
45 const int stepsPerRevolution = 2048;
46 Stepper myStepper = Stepper(stepsPerRevolution, 7, 12, 8, 13);
47
48 // Variables
49 long duration;
50 float distance;
51 void ultrasonic();
52 void blinkDistance();
53
54 void setup() {
55   pinMode(trigPin, OUTPUT);
56   pinMode(echoPin, INPUT);
57   pinMode(redID, OUTPUT);
58   pinMode(p_o, OUTPUT);

```

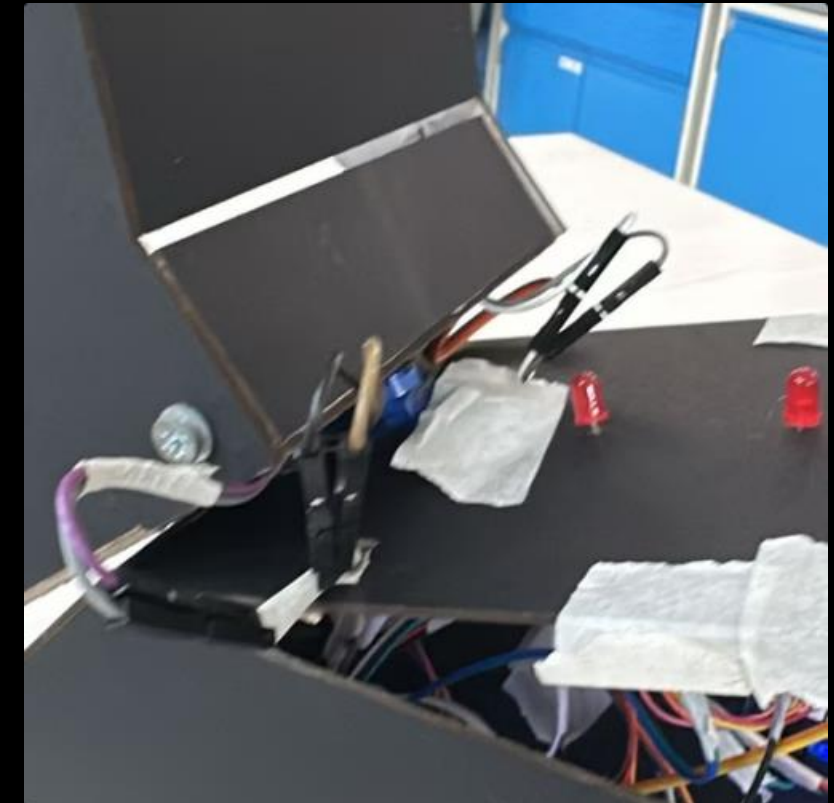
Programming

Arduino code was written to define the dinosaur's features.



Building the Circuit

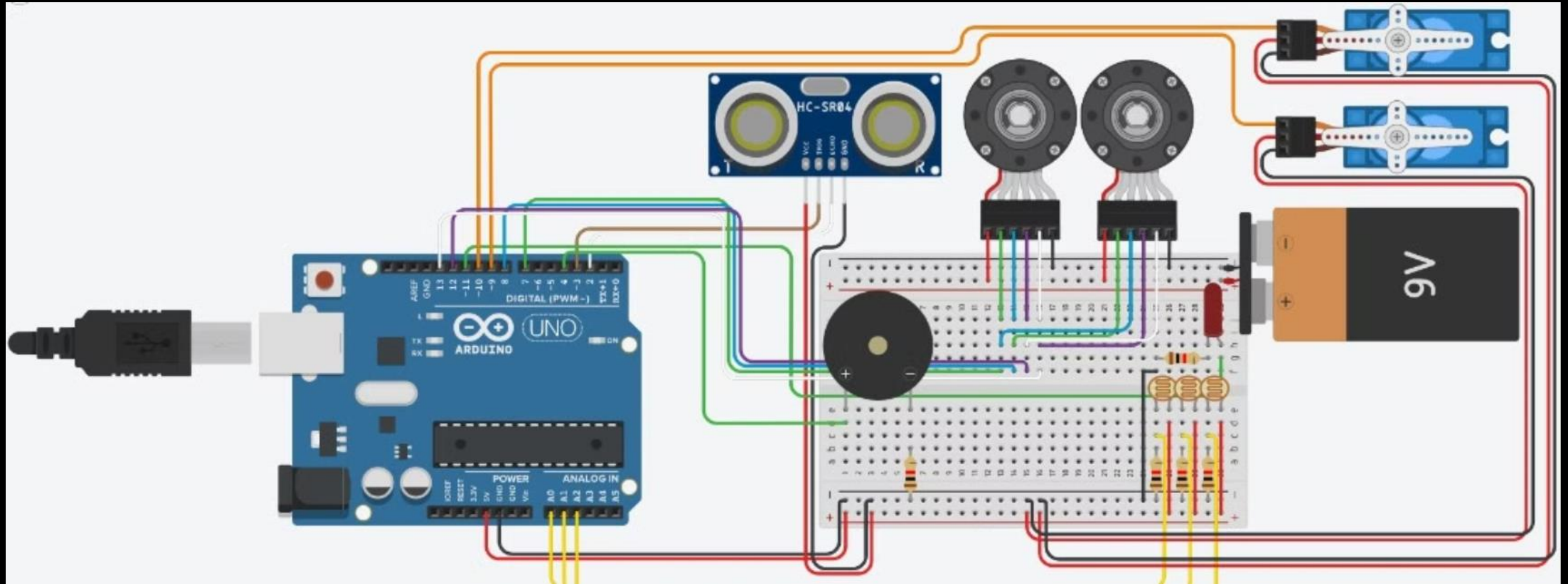
Arduino board was carefully wired to desire electronic components such as servo motors and sensors, creating a functional system.



Testing

Extensive testing was conducted to ensure smooth operation and seamless integration.

Full Circuit Diagram



Animating the Dinosaur: Servo/Stepper Movements and Sensors



Light Triggered Head Movement

When LDR detects light source ,servo motors controlling the dinosaur's neck turns the head



Sound Effects

Biting triggered sound effects, creating a more immersive experience and enhancing the dinosaur's presence.



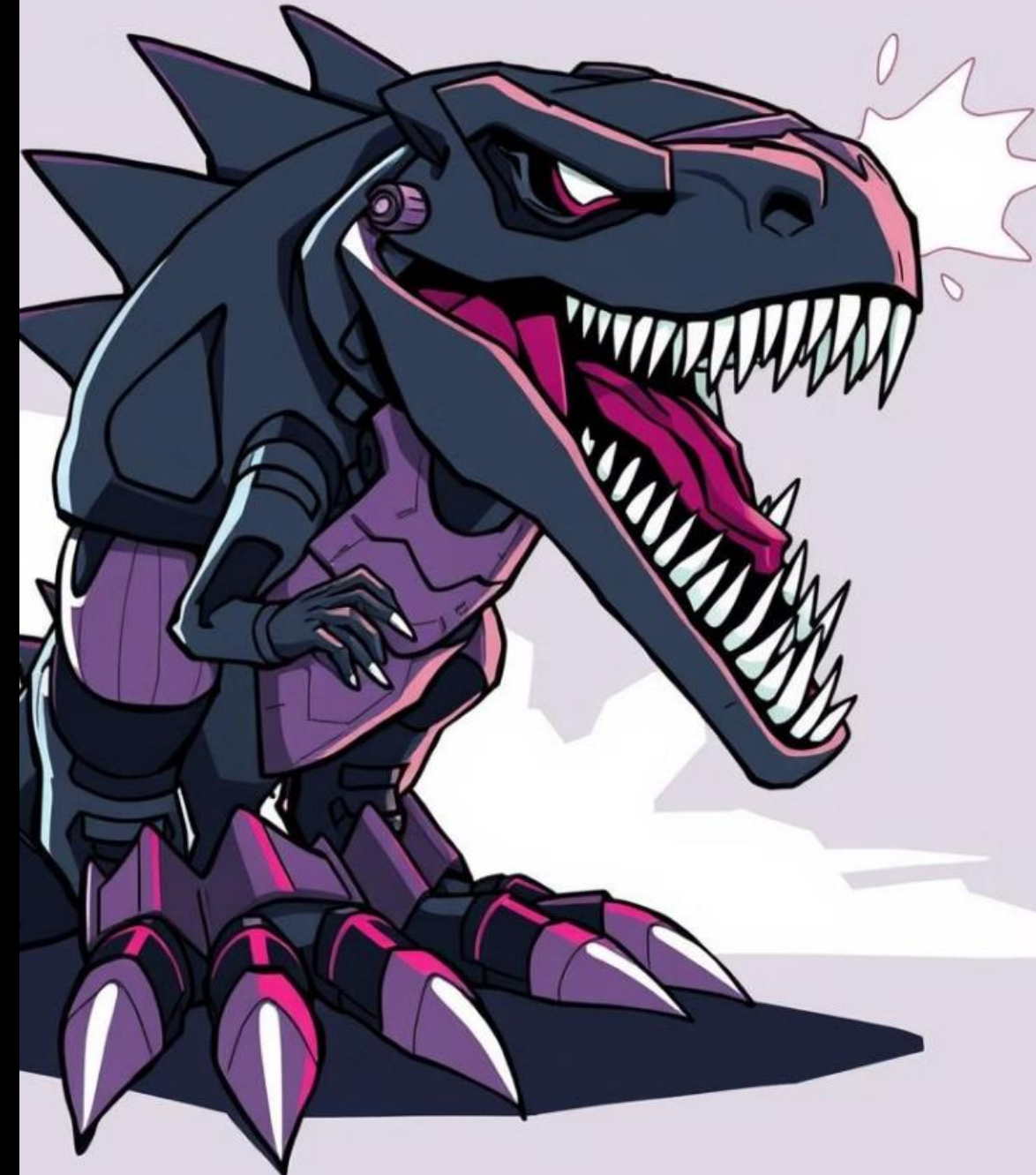
Mouth Movement

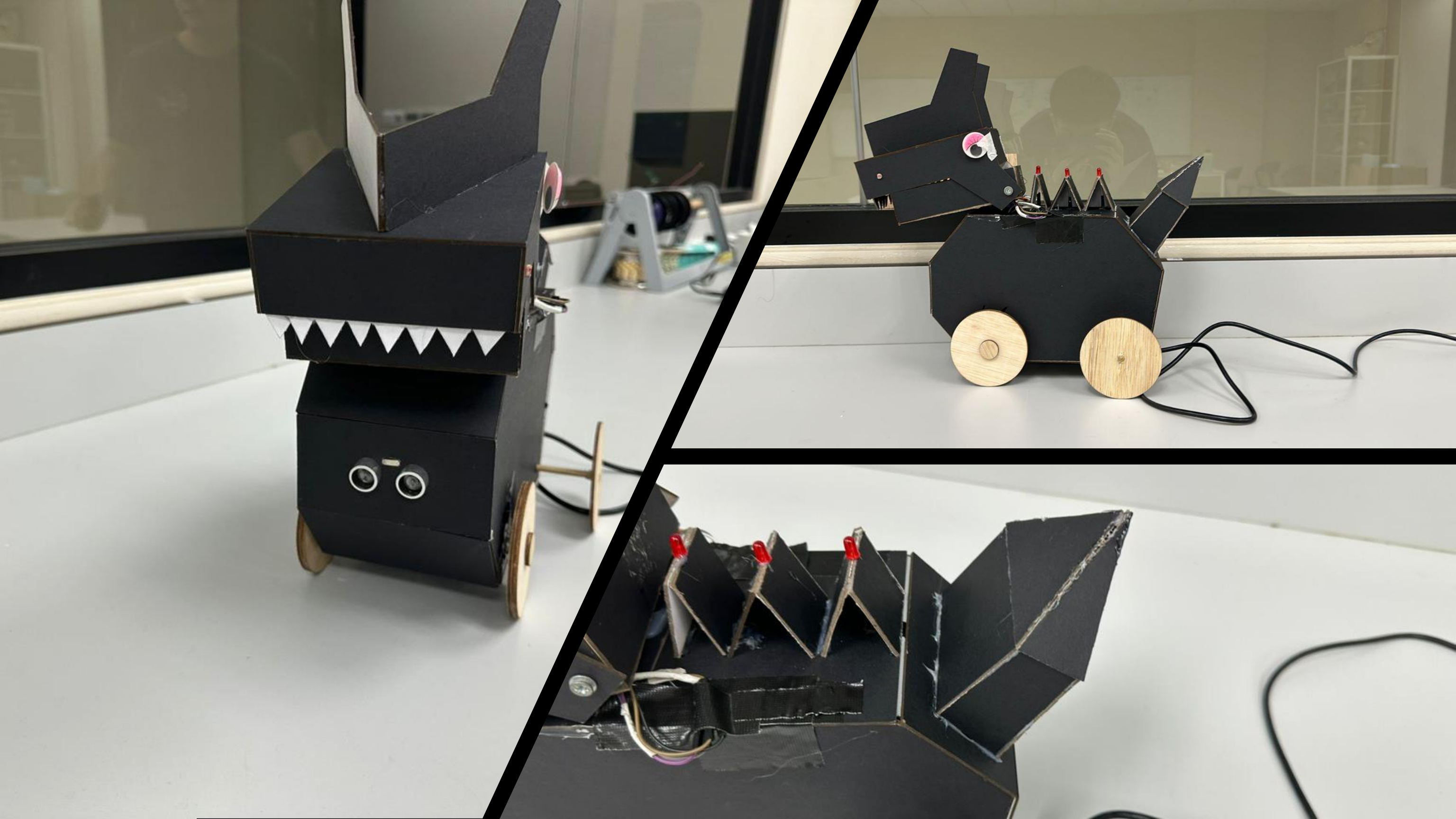
Servos enabled the dinosaur to close its mouth simulating a biting motion adding to its dynamic aggressive personality.



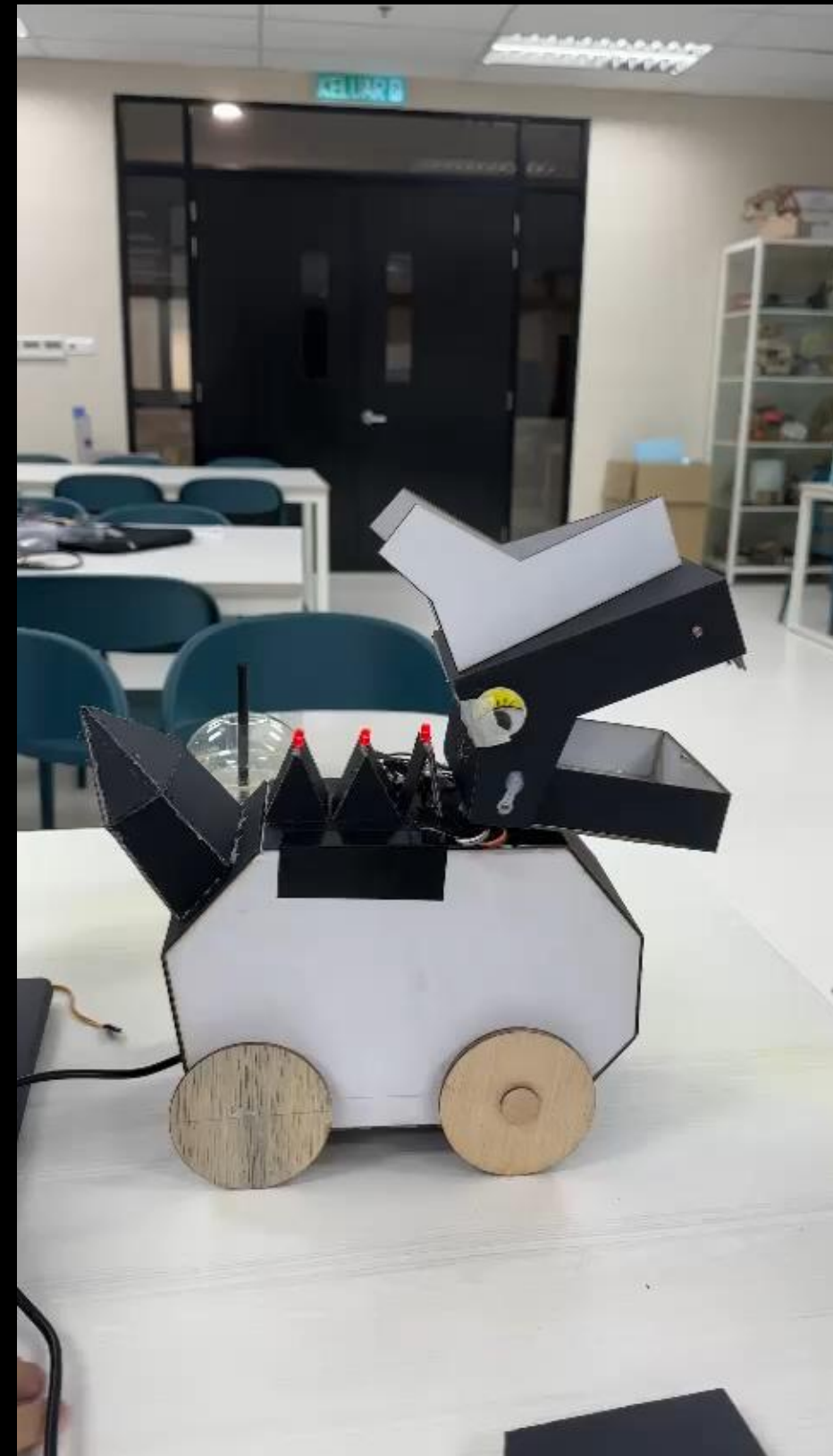
Body Movement

Body moves based on its fear and curiosity personality. When fearful, it moves backward to retreat, and when curious, it moves forward to explore, both controlled by the stepper motor in response to sensor inputs.





Final tests
ahead of
presentation
day.



Showcasing Rexduino's Capabilities

1

Dynamic personality-driven movements: retreating backward in fear and charging forward with aggressive intent.

2

Biting motion is enhanced with synchronized sound effects, delivering a realistic and captivating interactive experience.

3

Head tilts dynamically toward detected light sources, guided by the precision of its LDR sensor.





The Future of Arduino-Powered Robotics

1

Accessibility

Arduino's accessibility makes robotics accessible to a wider audience, encouraging innovation and creativity.

2

Education

Rexduino demonstrates the educational potential of robotics, inspiring future generations of engineers and inventors.

3

Entertainment

Rexduino shows how robotics can enhance entertainment, creating engaging and immersive experiences for all ages.